

Shranjeet Singh

501-333-4565 | shranjeetsingh03@gmail.com | [linkedin.com/in/sharanjeet-singh-](https://www.linkedin.com/in/sharanjeet-singh-) | professionalwebsite.vercel.app | US Citizen

EDUCATION

University of California, Riverside

June 2025

Bachelor of Science in Computer Science

Relevant Coursework: Software Engineering, Data Structures & Algorithms, Operating Systems, Theory of Automata, Compiler Design, Artificial Intelligence, Web Development, Data Science, Embedded Systems, Software Testing, Computer Networks

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Jan 2024 – June 2025

Center for Environmental Research & Technology, Riverside, CA

- Engineered a high-performance full-stack air quality research platform using React, Next.js, and TailwindCSS to display real-time data from **100+** sensors, cutting manual data compilation by **75%**.
- Implemented an interactive ArcGIS-powered map with AQI calculations and dynamic sensor popups, improving researcher efficiency through real-time visualization of pollution trends and reducing sensor data lookup time by **50%**.
- Optimized backend data transformation by restructuring QuantAQ API integration, reducing API response time by **40%** and enabling seamless data retrieval by eliminating **60%** of redundant requests via Next.js API routes.
- Developed real-time pollutant trend visualizations using VisX, displaying hourly sensor data per device and reducing anomaly detection time by **70%**.

Machine Learning Engineer Intern

Jun 2024 – Aug 2024

Bourns College Of Engineering, Riverside, CA

- Trained custom Wav2Vec2 deep learning model on **200+** multilingual audio recordings using transfer learning, achieving **40%** lower Word Error Rate for accented English compared to baseline models.
- Engineered comprehensive data preprocessing pipeline using PyDub and SoundFile libraries that converts diverse audio formats, creates train/validation/test splits, and generates training manifests for **100+** countries' worth of accented speech data.
- Developed benchmarking framework using HuggingFace Transformers that compares fine-tuned models against baseline and commercial services, achieving **30%** improvement in evaluation accuracy across diverse accent types.

Backend Developer Intern

Jun 2023 – Aug 2023

TRAPNSTUDIO, Riverside, CA

- Developed a multilingual AI chatbot backend using Node.js, **OpenAI** APIs, and Supabase, implementing conversation context tracking and language detection to improve response accuracy and accessibility for non-English-speaking users.
- Built a real-time music queue system with Node.js, WebSockets, and Redis, handling up to **50** concurrent users and reducing queue update latency by **15%** through event-driven architecture and efficient state synchronization.
- Implemented secure user authentication and role-based access control with Supabase Auth, enabling token-based session management and permission enforcement that reduced login failures by **20%**.

PROJECTS

AI Meeting Platform | TypeScript, Node.js, GraphQL, AWS Lambda, Supabase, OpenAI API

- Designed a serverless backend using **AWS Lambda** and **GraphQL (Apollo Server)** to process meeting audio and transcripts, storing structured summaries and action items in **Supabase**.
- Integrated **OpenAI API** for automatic summarization and action item extraction, reducing manual note-taking by **70%**.
- Implemented secure file handling and asynchronous processing pipelines with **AWS S3**, handling audio files up to **500MB**.

ML Traffic Enforcer | Python, PyTorch, FastAPI, Docker, Kubernetes, Redis

- Trained a **PyTorch** computer vision model to detect overspeeding and illegal lane changes, achieving **92%** accuracy.
- Deployed the model as a **FastAPI** microservice containerized with **Docker** and orchestrated on a **Kubernetes** cluster.
- Implemented **Redis caching** to reduce inference latency by **35%** and automated deployments using **GitHub Actions** CI/CD.

SKILLS

Languages: JavaScript, TypeScript, Python, C++, HTML, CSS

Frameworks: React, Next.js, Next Auth, TailwindCSS, Express.js, Node.js, Apollo Server

Tools/Databases: MongoDB, Firebase, Supabase, PostgreSQL, SQL, SQLite, Redis, Elasticsearch, Git, GitHub, Cypress, Postman

Cloud & DevOps: AWS (Lambda, S3, EC2), Docker, Kubernetes, GitHub Actions, Jenkins, Terraform

Other: Jira, Agile, Scrum Methodologies, GitHub Projects